INCH-POUND
MIL-I-24768/15
25 February 1992

SUPERSEDING
MIL-P-15035C
10 January 1962

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, COTTON-FABRIC-BASE, PHENOLIC-RESIN (FBI)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of a woven cotton fabric substrate impregnated and bonded with a phenolic-resin matrix processed to conform to this specification.

Intended use: The laminated material covered by this specification is primarily suitable for machining and mechanical applications where a fine edge detail and appearance is required. It possesses a high density and good moisture resistance. Its use for electrical applications is not recommended.

Forms: Sheets, molded tubes, and rods are recommended forms of laminated material covered by this specification.

Temperature indice: The laminated material covered by this specification has a mechanical strength test index of 85 and an electrical strength test index of 85 for thicknesses up to 0.062 inch, and a mechanical strength test index of 125 and an electrical strength test index of 115 for thicknesses greater than 0.062 inch.

Dimensions and part numbers:

Thickness of sheets: The thickness of laminated sheets, permissible variations, and the applicable part number shall be as specified in table I.

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TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inches)	Permissible variations (± inch)[3]
M24768/15-S-1	0.010	0.003
M24768/15-S-2	0.015	0.035
M24768/15-S-3	0.020	0.004
M24768/15-S-4	0.025	0.0045
M24768/15-S-5	0.031	0.005
M24768/15-S-6	0.047	0.0055
M24768/15-S-7	0.062	0.006
M24768/15-S-8	0.094	0.007
M24768/15-S-9	0.125	0.008
M24768/15-S-10	0.156	0.009
M24768/15-S-11	0.188	0.010
M24768/15-S-12	0.219	0.011
M24768/15-S-13	0.250	0.024
M24768/15-S-14	0.312	0.029
M24768/15-S-15	0.375	0.034
M24768/15-S-16	0.438	0.038
M24768/15-S-17	0.500	0.042
M24768/15-S-18	0.625	0.048
M24768/15-S-19	0.750	0.054
M24768/15-S-20	0.875	0.060
M24768/15-S-21	1.000	0.065
M24768/15-S-22	1.125	0.069
M24768/15-S-23	1.250	0.073
M24768/15-S-24	1.375	0.077
M24768/15-S-25	1.500	0.081
M24768/15-S-26	1.625	0.085
M24768/15-S-27	1.750	0.089
M24768/15-S-28	1.875	0.093
M24768/15-S-29	2.000	0.097
M24768/15-S-30	2.125	0.105
M24768/15-S-31	2.500	0.113
M24768/15-S-32	2.750	0.121
M24768/15-S-33	3.000	0.130
M24768/15-S-34	3.500	0.146
M24768/15-S-35	4.000	0.163
M24768/15-S-36	4.500	0.179
M24768/15-S-37	5.000	0.190
M24768/15-S-38	5.500	0.210
M24768/15-S-39	6.000	0.230
M24768/15-S-40	6.500	0.240
M24768/15-S-41	7.000	0.260
M24768/15-S-42	7.500	0.280
M24768/15-S-43	8.000	0.290
M24768/15-S-44	8.500	0.310
M24768/15-S-45	9.000	0.320
M24768/15-S-46	9.500	0.340
M24768/15-S-47	10.000	0.360
M24768/15-S-48	_[1]	_[2]

See footnotes on next page.

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[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet is: M24768/15-S-48-0.8125.
[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.
[3]Permissible variations for sheets greater than 0.219 inch thick shall be plus inch only.

Diameter of molded round tubes: The range of sizes for molded round tubes shall be as specified in table II. The inside diameter and outside diameter shall be included in the part number. An example of a part number for a molded round tube with an inside diameter of 0.125 inch and an outside diameter of 0.188 inch is: M24768/15-TMR-0.125/0.188. The wall thickness tolerances for round tubes with an inside diameter up to 3.875 inches shall be as specified in table III.

TABLE II. *Range of sizes for molded round tubes.*

Inside diameter (inches)		Outside diameter (inches)		Wall thickness (inch)		Maximum ratio of wall thickness to inside diameter
Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	
0.125	3.875	0.188	4.000	0.031	1.000	1/2

TABLE III. *Molded round tube wall thickness tolerances.*

Wall thickness (inches)	Tolerances (± inch)		
	Inside diameter (inches)		
	0.125–0.250	0.251–0.500	0.501–Max.
0.031 to 0.061	0.008	0.008	0.008
0.062 to 0.125	0.011	0.011	0.011
0.126 to 0.250		0.015	0.011
0.251 to 0.500			0.013

Diameter of rods: The range of sizes and tolerances for rods shall be as specified in table IV. The outside diameter of the rod shall be included in the part number. An example of a part number for a rod with an outside diameter of 0.188 inch is: M24768-15-R-0.188.

TABLE IV. *Size and tolerances for rods.*

Diameter (Inches)	Tolerances (± inch)
0.125 to 1.999	0.005
2.000 to 4.000	0.008

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Performance requirements: Performance requirements for laminated materials shall be as specified in tables V, VI, VII, and VIII.

The following performance requirements do not apply to this plastic laminate:

a. Dissipation factor
b. Dielectric constant
c. Surface resistance
d. Volume resistivity
e. Arc resistance
f. Flammability
g. Tracking resistance
h. Thermal endurance
i. Silicone content
j. Dielectric strength.

Testing: The laminated material covered by this specification shall be tested under first article inspection requirements.

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TABLE V. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Dielectric breakdown voltage[2]	A	4.6.4	Min. kV	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	10.0
Dielectric constant	A D-24/23	4.6.6	Max.										
Dissipation factor	A D-24/23	4.6.6	Max.										
Volume resistivity	C-96/35/90	4.6.7	Min. megohm cm										
Surface resistance	C-96/35/90	4.6.8	Min. megohm										
Arc resistance	D-48/50	4.6.9	Min. sec.										
Tracking resistance	A	4.6.10	Min. minutes										
Impact strength[2]			Min. ft-lbs. per inch										
Lengthwise:	E-48/50	4.6.11					1.35	1.35	1.35	1.35	1.35	1.35	1.35
Crosswise:							1.10	1.10	1.10	1.10	1.10	1.10	1.10
Flexural strength													
Lengthwise:	A	4.6.12	Min. psi	16,500	16,500	16,500	16,500	16,500	16,500	15,500	15,500	15,000	15,000
Crosswise:				14,500	14,500	14,500	14,500	14,500	14,500	14,000	14,000	13,500	13,500

See footnotes at end of table.

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TABLE V. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Thermal endurance													
Flexural strength:		4.6.13	Min. psi										
Bonding strength[2]	A		Min. pounds							1,600	1,600	1,600	1,600
	D-48/50	4.6.15								1,500	1,500	1,500	1,500
Water absorption	D_1-24/23	4.6.17	Max. percent	6.00	2.50	1.90	1.60	1.30	1.10	0.90	0.75	0.70	0.70
Silicone content	E-168/185	4.6.18	Max. ppm										
Flame resistance													
Ignition time:	A	4.6.19	Min. sec.										
Burning time:			Max. sec.										

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.
[2]Maximum thickness tested shall be 2.000 inches.

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TABLE VI. *Performance requirements for molded tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inch)							
					0.031–0.061	0.062–0.093	0.094–0.124	0.125–0.187	0.188–0.249	0.250–0.499	0.500–0.999	1.000
Compressive strength	A	4.6.14	Min. psi	0.250 to 3.875		18,000	18,000	18,000	18,000	18,000	18,000	18,000
Specific gravity	A	4.6.16	Min.	0.125 to 3.875	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25
Water absorption	D_1-24/23	4.6.17	Max. percent	0.125 to 3.875	6.5	3.5	2.2	1.8	1.6	1.6	1.5	1.4

TABLE VII. *Dielectric strength requirements for molded tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inch)			
					0.062	0.063–0.125	0.126–0.250	0.251–0.500
Dielectric strength	A	4.6.5	Min. volts per mil	0.125–3.875				

TABLE VIII. *Performance requirements for rods.*

Requirement	Cond.	Test para.	Unit	Diameter (inches)				
				0.188–0.249	0.250–0.499	0.500–0.999	1.000	1.001–2.000
Flexural strength	A	4.6.12	Min. psi	16,000	16,000	16,000	16,000	
Compressive strength	A	4.6.14	Min. psi	19,000	19,000	19,000	19,000	19,000
Specific gravity	A	4.6.16	Min.	1.28	1.28	1.28	1.28	1.28
Water absorption	D_1-24/23	4.6.17	Max. percent	2.5	2.0	1.5	1.2	1.2

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Custodians:
Army – ER
Navy – SH
Air Force – 20

Review Activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User Activities:
Army – ME
Navy – MC, AS

Preparing Activity:
Navy – SH
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